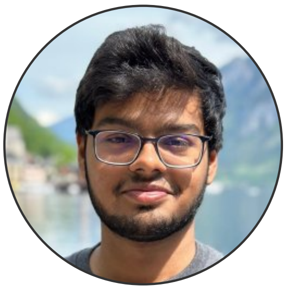

Ashmit Khandelwal

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Education

Carnegie Mellon University — Pittsburgh, PA

M.S. Machine Learning — Incoming, Fall 2026

Birla Institute of Technology and Science, Pilani — India

B.E. Computer Science, Minor in Data Science — 2020 - 24

GPA: 9.5/10.0

Work Experience

Research Fellow - Microsoft Research — Bangalore, India

Advisors: Nagarajan Natarajan, Amit Sharma — Jul 2024 - June 2026

- Developed interwhen, a test-time verification framework to asynchronously monitor and steer LLM agent trajectories. Spans code generation, arithmetic, logical, spatial, and agentic settings.
- Formally defined Deep Research (DR), a previously under-defined class of LLMs for large-scale search and reasoning. Introduced LiveDRBench, one of the first benchmarks to evaluate DR, highlighting performance gaps in DR systems from OpenAI, Perplexity, and Gemini.
- Studied robustness of Direct Preference Optimization (DPO) under noisy LLM labels produced by prompt-based and finetuned verifiers.

Research Intern - Adobe — Noida, India

Advisors: Yaman Kumar (Adobe), Changyou Chen (SUNY Buffalo) — Summer 2023

- Modeled human behavior from YouTube videos to predict rewatchability, engagement metrics, and audience sentiment at scale across diverse content domains.
- Designed and trained multimodal LLM architectures that integrated video and textual representations, achieving significant performance gains and surpassing GPT-4 on multiple behavior prediction tasks.
- Built pipelines for video, audio, and text scraping and processing, automating the collection of terabytes of web data.

Publications

* = equal contribution

Large Content And Behavior Models To Understand, Simulate, And Optimize Content And Behavior — Spotlight at ICLR

A Khandelwal*, A Agrawal*, A Bhattacharyya*, YK Singla*, et al. — 2024

Large Content and Behavior Models (LCBMs) reintroduce behavior tokens into LLM training data to simulate and explain human audience behavior in response to different forms of content. These models generalize across content types and adapt to diverse behavior domains, demonstrated on the newly introduced Content Behavior Corpus (CBC).

Characterizing Deep Research: A Benchmark and Formal Definition — ICLR

A Java*, A Khandelwal*, S Midigeshi*, et. al. — 2026

Deep Research (DR) systems lacked a formal definition despite the emergence of several proprietary and open-source models. This work established a concrete definition of DR, introduced LiveDRBench as one of the first DR benchmarks along with a recursive evaluation framework. Highlighted performance gaps in DR systems from OpenAI, Perplexity, and Gemini.

interwhen: A Generalizable Framework for Verifiable Reasoning with Test-time Monitors — VerifyAI at ICLR

V. Bhat*, P. Chanda*, V. Ekbote*, A Khandelwal, M. Swaroop, et. al. — 2026

Interwhen is a test-time verification framework for reasoning models to asynchronously verify intermediate reasoning. We improve reasoning soundness across code generation, logical reasoning, and agentic settings, without the high token overhead of standard test-time scaling methods.

Benchmarking VLMs' Reasoning About Persuasive Atypical Images WACV

S Malakouti*, A Aghazadeh*, A Khandelwal, A Kovashka 2025

Benchmarked vision-language models on inferring the message in rhetorical and atypical imagery used in advertisements, through three novel tasks. Found that VLMs struggle with reasoning about atypicality, but simple methods enable atypicality-aware verbalization and improve rhetorical image understanding.

Research Explorations

Simulating the Unpredictability of Human Society — Adobe

Advisors: Yaman Kumar, Veeky Baths — Fall 2023

- Measured monthly Jensen–Shannon divergence in English Wikipedia visit and edit distributions, revealing a steady rise in entropy and growing randomness in collective information.
- Simulated event propagation on a graph-based SIR model to study how new events increase uncertainty and drive the observed rise in entropy.

Semi-Supervised Segmentation and VQA on Aerial Flood Images — BITS Pilani

Advisor: Sravan Danda — 2022 - 23

- Designed a semi-supervised segmentation and graph-based VQA system for the FloodNet challenge, combining CutMix and Cross Pseudo Supervision.
- The VQA module uses geodesic dilation and morphological operations on segmentation maps, followed by reasoning over 4-adjacency graphs to count connected components.

Teaching Experience

Operating Systems — BITS Pilani

Teaching Assistant | ~250 Students — Fall 2023

- Undergraduate course on operating system principles, including process management, memory management, and semaphore mechanisms.
- Responsibilities included lab sessions, lab test preparation, and student mentoring.

Introduction to ML and DL — BITS Pilani

Instructor | ~40 Students — Spring 2023

- Student run course on foundational ML/DL concepts and algorithms, and programming with python, numpy, and pytorch.
- Responsibilities included curriculum design, lecture delivery, assignment creation, and student mentoring.

Object Oriented Programming — BITS Pilani

Teaching Assistant | ~250 Students — Fall 2022

- Undergraduate course introducing object-oriented programming concepts using Java.
- Responsibilities included lab sessions, lab test preparation, student mentoring.